

Support vector machines: kernels for nonlinearity

ECMM422

Fabrizio Costa

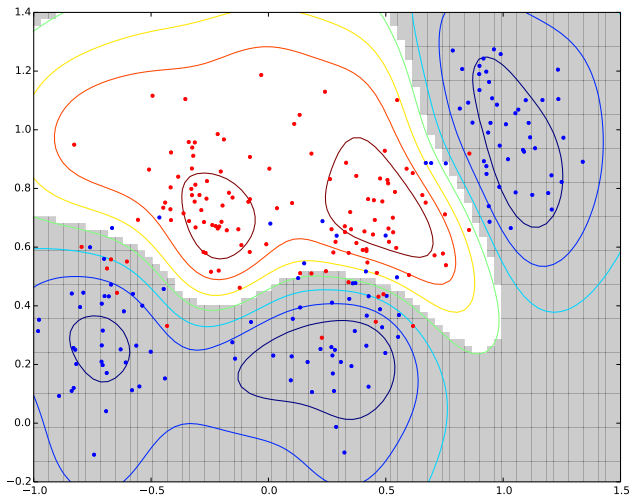
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1 Non Linear Large Margin Classifiers

2 The Curse of Dimensionality

Large margin classifiers on steroids



- ▶ Nonlinearly map the data to some high-dimensional space
- ▶ Use a large margin classifier to separate the classes in the high-dimensional space

Mapping to higher dimensions

Nonlinear mapping of data to higher dimensions may allow it to be linearly separated

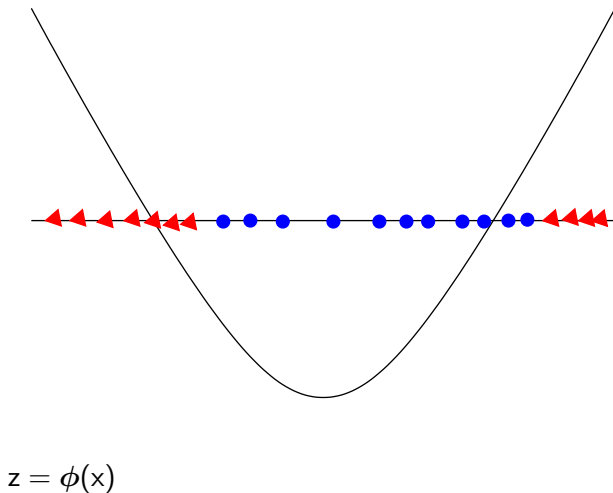
In the example, introducing a second dimension allows the one-dimensional data to be separated by a line.



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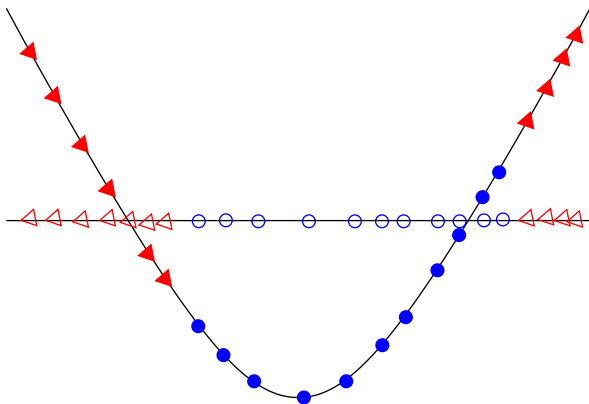
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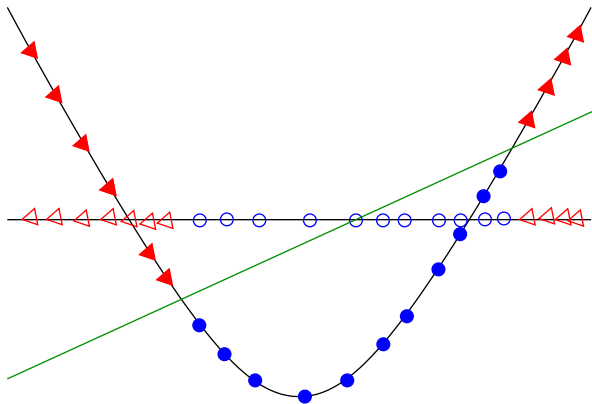
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New coordinates: $x_n \mapsto z_n = \phi(x_n)$

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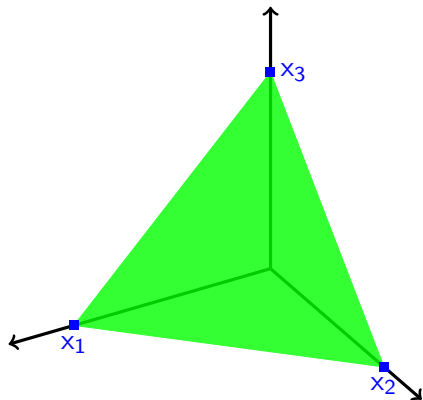
Cover's theorem

Cover's theorem

A dataset mapped to a higher dimensional space is more likely to be linearly separable

Proof

Map $\{x_n\}_{n=1}^N$ to the vertices of an $N - 1$ -dimensional simplex (embedded in $\mathbb{R}^N$). Then every partition of the points into two sets is separable by a linear hyperplane.



Mapping to higher dimensions

Cover's theorem: a dataset mapped to a higher dimensional space is more likely to be separable

Map data x to a higher dimensional feature space with

$$z = \phi(x)$$

for some nonlinear *basis function* $\phi(\cdot)$

- ▶ $z_n = \phi(x_n)$ is a surrogate for x_n in the new high dimensional feature space
- ▶ Dimension of z might be infinite

Classify z with large margin classifiers in the high dimensional feature space

Finding the maximum margin hyperplane can be done using dot products between the z_n

Lagrangian dual function with feature map ϕ

$$\max_{\alpha} \sum_n \alpha_n - \frac{1}{2} \sum_{n,m} t_n t_m \alpha_n \alpha_m \langle \phi(x_n), \phi(x_m) \rangle$$

$$\text{such that } 0 \leq \alpha_n \leq C, \quad \sum_n \alpha_n t_n = 0$$

Discriminant function

$$y(x) = \sum_{n \in SV} \alpha_n t_n \langle \phi(x_n), \phi(x) \rangle + b$$

To determine b , we can find some index i such that $0 < \alpha_i < C$, so that $\phi(\mathbf{x}_i)$ lies on the boundary of the margin in the transformed space, and then solve:

$$b = t_i - \mathbf{w}^T \phi(\mathbf{x}_i) = t_i - \left[\sum_n \alpha_n t_n \phi(\mathbf{x}_n) \cdot \phi(\mathbf{x}_i) \right].$$

- ▶ Nonlinear SVM only needs dot products between training and test data points

$$K = (\langle \phi(x_n), \phi(x_m) \rangle)_{m,n}^N \quad K_x = (\langle \phi(x_n), \phi(x) \rangle)_n^N$$

- ▶ Solving the dual QP problem for N Lagrange multipliers requires $\mathcal{O}(N^3)$ time, *independent of the dimension of high-dimensional feature space*

From feature maps to kernels

- ▶ A feature map is a function from the original data space to a high-dimensional *feature space*:

$$\phi : \mathbb{R}^d \mapsto \mathbb{R}^D$$

- ▶ Feature space may be any *Hilbert space*,¹ $\mathcal{W}$.
- ▶ A feature map $\phi(\cdot)$ is associated with a kernel function

$$K(x, y) = \langle \phi(x), \phi(y) \rangle \quad x, y \in \mathbb{R}^d$$

Example

- ▶ If $\phi(x) = (x_1^2, x_2^2, \sqrt{2}x_1x_2)$ for $x \in \mathbb{R}^2$
- ▶ Kernel function is

$$K(x, y) = (x^T y)^2$$

¹Hilbert spaces are the infinite dimensional analogue of Euclidean space. They are equipped with an *inner product* so lengths of “vectors” and angles between them are defined.

Positive semi-definite kernels

A function $K : \mathbb{R}^d \times \mathbb{R}^d \mapsto \mathbb{R}$ is positive semi-definite (psd) if it is symmetric and the matrix $(K(\mathbf{x}_i, \mathbf{x}_j))_{i,j=1,\dots,M}$ is psd for any M and $\mathbf{x}_1, \dots, \mathbf{x}_M \in \mathbb{R}^d$

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Mercer's Theorem

K is positive semi-definite if and only if

$$K(\mathbf{x}, \mathbf{y}) = \langle \phi(\mathbf{x}), \phi(\mathbf{y}) \rangle \quad \mathbf{x}, \mathbf{y} \in \mathbb{R}^d$$

for some feature map

$$\phi : \mathbb{R}^d \mapsto \mathcal{W}$$

But generally the precise feature map $\phi(\cdot)$ is unknown

Kernels

Linear kernel

$$K(x, y) = x^T y$$

Polynomial kernel

$$K(x, y) = (1 + x^T y)^r$$

Radial Basis Function (aka RBF, aka Gaussian)

$$K(x, y) = \exp \{-\gamma^2 \|x - y\|^2\}$$

Parameter γ^2 is a precision (inverse variance) controlling the strength of interaction of one observation with another

Others

Many, many more. Can be tailored for particular applications to reflect similarity of observations

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Properties

If K_1 and K_2 are kernels then

- ▶ $\alpha K_1 + \beta K_2$ is a kernel for $\alpha, \beta \geq 0$
- ▶ $K_1 \cdot K_2$ is a kernel

RBF kernel implies infinite dimensional feature space

Remember the Taylor series expansion of a function f at a

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n,$$

for the exponential function at $a = 0$ the expansion becomes:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

RBF kernel implies infinite dimensional feature space

Let $x \in \mathbb{R}$ and $\gamma > 0$.

$$e^{-\gamma\|x-y\|^2} = e^{-\gamma(x-y)^2} = e^{-\gamma x^2 + 2\gamma xy - \gamma y^2} = e^{-\gamma x^2 - \gamma y^2} e^{2\gamma xy}$$

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with

$$\phi(x) = e^{-\gamma x^2} \left[1, \sqrt{\frac{2\gamma}{1!}} x, \sqrt{\frac{(2\gamma)^2}{2!}} x^2, \sqrt{\frac{(2\gamma)^3}{3!}} x^3, \dots \right]^T$$

SVM algorithm

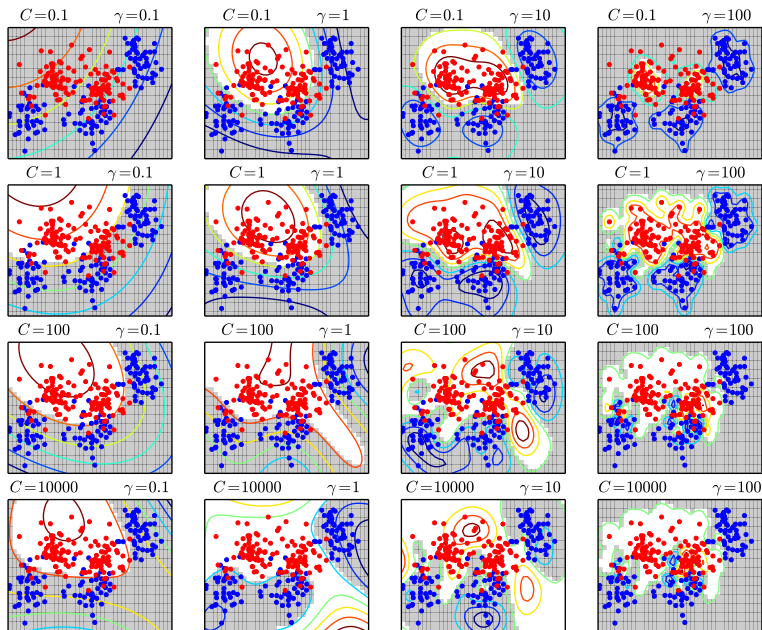
- 1 Choose a kernel function
 - ▶ Gaussian or polynomial common, but may need particular kernels for particular problems reflecting similarities
- 2 Choose a regularisation constant C
 - ▶ E.g. by using cross-validation for C and any other parameters (e.g., γ)
 - ▶ or set $C \rightarrow \infty$ for hard margin solution
- 3 Solve the dual quadratic program for α_n
- 4 Construct discriminant function from support vectors

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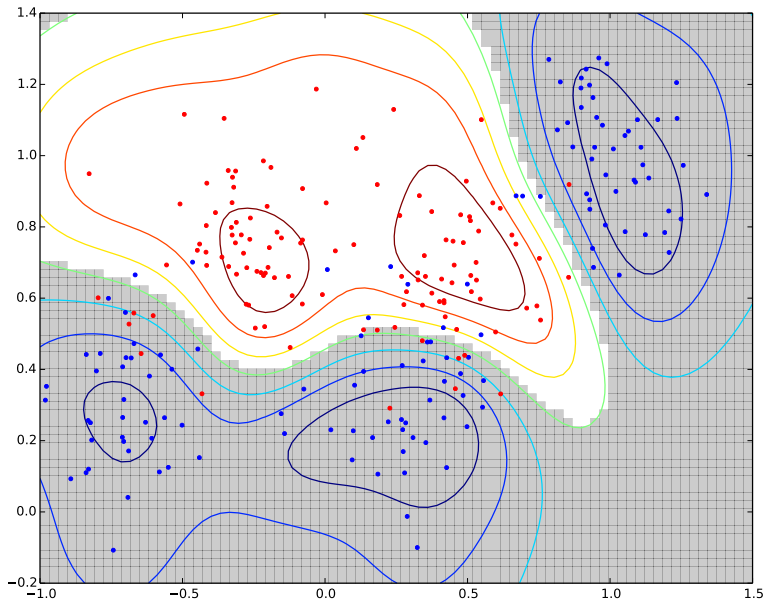
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Posterior probabilities: Platt (1999)

- ▶ Map the learned discriminant to $(0, 1)$ with *logistic function* $1/(1 + e^{-ay(x)+b})$.
- ▶ Estimate a and b using maximum likelihood (cross entropy) and cross validation to prevent over-fitting.



Best SVM: $\gamma = 10$, $C = 1$



Soft margin SVM

$$\begin{aligned} \min_{w,b} \quad & \frac{1}{2} \|w\|^2 + C \sum_n \xi_n \\ \text{s.t.} \quad & t_n(w^T x_n + b) \geq 1 - \xi_n \\ & \xi_n \geq 0 \text{ for any } n \end{aligned}$$

- Rewrite constraint as

$$\xi_n \geq 1 - t_n(w^T x_n + b)$$

putting it together with $\xi_n \geq 0$ we get:

$$\xi_n \geq \max(0, 1 - t_n(w^T x_n + b)) \equiv [1 - t_n(w^T x_n + b)]_+$$

Soft margin SVM

$$\min_{w,b} \quad \frac{1}{2} \underbrace{\|w\|^2}_{\text{Regularisation}} + C \underbrace{\sum_n [1 - t_n(w^T x_n + b)]_+}_{\text{Data/empirical error}}$$

General regularisation network

$$E_{\lambda}(w) = \underbrace{\sum_{n=1}^N L(t_n, \langle w, \phi(x_n) \rangle)}_{\text{Loss/Data error}} + \lambda \underbrace{\langle w, w \rangle}_{\text{Regularisation}}$$

where $L : \mathbb{R} \times \mathbb{R} \mapsto \mathbb{R}$ is a *loss function* measuring how much we care about mismatching the prediction to the data

Regression $L(t, y(x)) = (t - y(x))^2$

Logistic regression $L(t, y(x)) = \log(1 + e^{-ty(x)})$

SVM with hinge loss $L(t, y(x)) = [1 - ty(x)]_+$

1 Non Linear Large Margin Classifiers

2 The Curse of Dimensionality

The Dimensionality Issue

- ▶ When we perform the bias-variance analysis of k-nearest-neighbor: seems that when $n \rightarrow \text{LARGE}$ the model can approximate well $\Rightarrow$ low bias and vanishing variance
 - ▶ We just need to average the information of the k-nearest neighbors!
- ▶ However, we need many neighbors near any given x
 - ▶ In high dimensional spaces this assumption is not trivial

Curse of Dimensionality

- ▶ Consider inputs uniformly distributed in a p -dimensional hypercube
- ▶ Consider a hyper-cubical neighborhood to capture a fraction r of inputs (or equivalently of the unit volume)
- ▶ The edge length of this neighborhood is therefore

$$l_p(r) = r^{\frac{1}{p}}$$

{so that the volume is the desired fraction $l_p(r)^p = r$ }

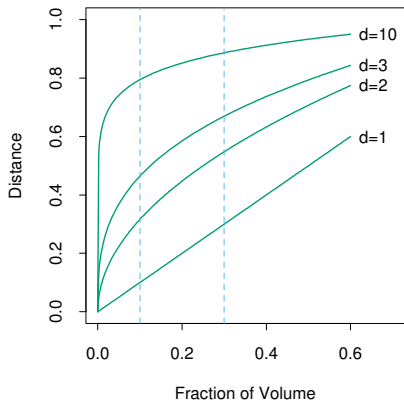
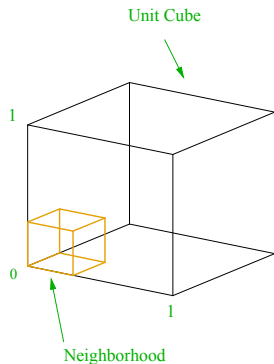
- ▶ Already in 10 dimensions

$$I_{10}(0.01) = 0.63$$

$$I_{10}(0.1) = 0.80$$

- ▶ This means that if we want to cover 1% of the data to form a reasonable average, we have to extend the neighborhood to more than half of the entire data space!
- ▶ Not to mention 10% of the data (80% of the variable range)

Curse of Dimensionality



Problems with High Dimensionality

- ▶ This shows that the neighborhoods are not *local* anymore!
- ▶ We cannot just try to have very small r , since this would imply few data points to average and therefore a high variance
- ▶ Another problem is that all points are close to the *edge* of the sample
- ▶ This implies that we need to *extrapolate* (more difficult) rather than interpolate (easier)

- ▶ Consider N data points uniformly distributed in a p -dimensional unit ball centered at the origin
- ▶ Consider the neighborhood of the origin
- ▶ The median distance of the closest point from the origin is:

$$d(p, N) = \left(1 - \frac{1}{2}\right)^{\frac{1}{p}}$$

- ▶ For $N = 500$ and $p = 10$ we have $d(p, N) \approx 0.52$
- ▶ That is:
 - 1 the closest point is likely to be more than half way to the boundary
 - 2 most points are closer to the boundary of the sample than to any other data point

Edge effect II

Another way to see the edge effect is:

- ▶ Consider a p -dimensional unit ball centered at the origin
- ▶ We want to compute the fraction of volume that lies between radius $R = 1 - \epsilon$ and $R = 1$
- ▶ The volume of a sphere is proportional to R^p :

$$V_p(R) = K_p R^p$$

where K_p depends only on p

- ▶ The volume fraction is therefore:

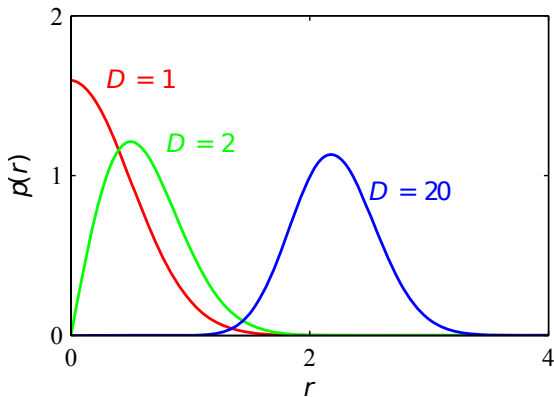
$$\frac{V_p(1) - V_p(1 - \epsilon)}{V_p(1)} = 1 - (1 - \epsilon)^p$$

- ▶ which for high values of p tends to 1 even for small ϵ
- ▶ I.e. most of the volume of a sphere is concentrated in a thin shell near the surface

Edge effect III

Another way to see the edge effect is:

- ▶ Consider the behavior of a Gaussian distribution for high p
- ▶ Transforming from Cartesian to polar coordinates, it can be shown that most of the probability mass $p(R)\delta R$ tends to be concentrated in a thin shell **far from the origin !**



A further problem in high dimension is the sparsity of data sets

- ▶ Given N data points uniformly distributed in a hyper cube of dimension p
- ▶ ..the average density is proportional to $N^{\frac{1}{p}}$
- ▶ If we say that with $N = 100$ we have a reasonable sample in 1 dimension
- ▶ ...then in order to achieve the same density with $p = 10$ we need 100^{10} data points!
- ▶ $\Rightarrow$ in high dimensional spaces any reasonable data sets can only **sparsely** populate the input space!

High Dimensional Trade-Off

- ▶ In order to fight the curse of dimensionality, one can constrain the complexity of the hypothesized data model
- ▶ The idea is to assume some regular/simple behavior in the neighborhood of data points and impose it by construction in the model

High Dimensional Trade-Off

- ▶ The strength of the constraint is related to the neighborhood size: the larger the size of the neighborhood the more sensitive is the solution to the choice of the constraint (the implicit bias)
- ▶ On the one end of the spectrum, there are linear models that have degenerate neighborhoods and take into account all points (but high bias)
- ▶ At the other end, there are non-parametric models (like the k -nearest neighbor) that can have arbitrary small isotropic neighborhoods

High Dimensional Trade-Off

- ▶ Any method that produces locally varying functions in **small isotropic neighborhoods** will have problems in high dimensions
The problem will likely manifest in the form of a high variance error component
- ▶ Methods that overcame the curse of dimensionality have an implicit or adaptive metric for measuring the neighborhood, so that the neighborhood is **not small in all directions**

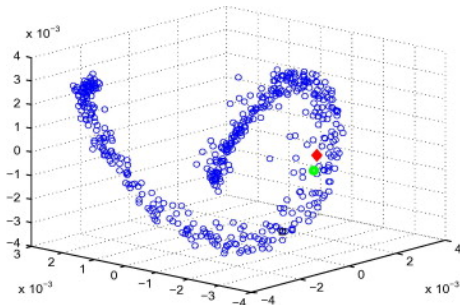
High Dimensional Fight

In order to have any hope in the fight against the curse of dimensionality, some of these conditions must be met:

- ▶ The number of *bumps* of the true function must be low (i.e. not scale exponentially with the dimensionality)
- ▶ The *degrees of freedom* of the true function must be low (possibly independent of the p -dimensionality)
in other words, data should lay in a **low dimensional manifold**
- ▶ A special case is when the state of nature depends on a small number of features

Low Dimensional Manifold

- ▶ A manifold of dimension n is a topological space that near each point resembles p -dimensional Euclidean space
- ▶ In other terms a manifold is a space for which there exist a chart/map that puts points in correspondence with a line or a plane or a cube, etc.
- ▶ Note that the correspondence can be non-linear
- ▶ Below, an approximately 1-dimensional manifold embedded in a 3D space



Low Dimensional Manifold

- ▶ It is not an unreasonable assumption to require that data should live in some low-dimensionality manifold
- ▶ Consider the image of a hand-written digit (say 3) as a point in a high dimensional space
- ▶ i.e. instead of considering a matrix 32×32 of pixels, consider each pixel as a feature by stacking all columns on top of each other to obtain a vector in $p = 1024$
- ▶ Consider the set of all images obtained by shifting the 3 horizontally



Low Dimensional Manifold

- ▶ It seems intuitive that the features of these instances are highly constrained
- ▶ In fact, they have only 1 degree of freedom
- ▶ I.e. in a vector space with 1024 dimensions, all the points lie on a *single line*
- ▶ Even when we increase the resolution (say to obtain a grid of 256x256) the manifold intrinsic dimension remains the same



Low Dimensional Manifold

- ▶ Even when the data set is not so trivially constructed, we can still believe the manifold assumption to hold
- ▶ There can be *traits* that correspond to the notion of degrees of freedom
- ▶ e.g. the degree of tilt, the length of the terminal curl, etc



- ▶ Note however that in general the manifold will not be *linear*

Further reading

- ▶ Bernard Scholkopf: *Learning with Kernels: Support Vector Machines, Regularization, Optimization and Beyond*, Springer, 2002.
- ▶ Nello Cristianni and John Shawe-Taylor: *An Introduction to Support Vector Machines and Other Kernel-based Learning Methods*, Cambridge University Press, 2000.
- ▶ Zeng ZQ, Yu HB, Xu HR, Xie YQ, Gao J. *Fast training Support Vector Machines using parallel sequential minimal optimization*. In: Intelligent System and Knowledge Engineering, 2008. ISKE 2008. 3rd International Conference on 2008 Nov 17 (Vol. 1, pp. 997-1001). IEEE.
- ▶ Chang CC, Lin CJ. *LIBSVM: a library for support vector machines*. ACM Transactions on Intelligent Systems and Technology (TIST). 2011 Apr 1;2(3):27.